SIMLOCK SYSTEM TESTING LABS

UNIFIED MASTER TECHNICAL SPECIFICATION & COMPLIANCE OPERATIONS

Unified Version 5.0 — Technical Reference, Interconnect Pinouts, & Execution Guide

This comprehensive system validation manual merges and re-specifies the complete logical, hardware, and software structures of the distributed SIMLOCK testing platform. Combining low-level hardware-pin matrix rationale, serial UART handshaking timing parameters, local web portal configurations, and remote fleet-wide cloud dashboard telemetry pipelines, this unified guide represents the ultimate integration specification for hardware development and software QA operations.

Metadata Specification Parameter	Assigned System Value / Reference
System Reference Name	SIMLOCK Distributed Multi-Board Test Platform
Target Devices Under Test	DL MCU Doorlock Controller & Camera MCU SPI Image Module
Local Agent Controller	Waveshare ESP32-C6-WROOM-1-N8 (8MB Flash Model)
Central Communication Broker	Central MQTT Broker (mqtt.ebizco.com.au / TLS Port 8883)
Unified Document Level	Level 1 Protocol Contract & Level 2 Operational Integration Guide
System Document Release	Version 5.0 (Reconstructed from v2.2 Technical Specs)
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CHAPTER 1: INTRODUCTION, SCOPE & DEFINITIONS

1.1 Purpose

This manual specifies the operational and structural properties of the distributed SIMLOCK test platform. This document details the exact hardware interface logic, serial line configurations, on-device Wi-Fi provisioning pathways, local browser-based execution logs, remote cloud backend reporting schemas, and complete portable test suite formats. This manual acts as a dual protocol specification (Level 1) and field execution manual (Level 2) for QA validation engineers, firmware developers, and operations supervisors.

1.2 Scope & System Coverage Boundaries

The scope of this technical reference covers the physical 5-wire serial interface, active-low line handshaking logic, SPI-based secondary peripheral mapping, local network access routines, on-bench captive portal wireframes, in-flight test result structures, and distributed client-broker publish topics. This document explicitly excludes physical motor actuator mechanics, internal door-lock MCU silicon structures, and secondary network routing firewalls beyond the designated central telemetry server.

1.3 Unified Definitions & Acronym Glossaries

Acronym / Term	Standard Functional System Definition
N8 Controller	ESP32-C6-DevKitC-1-N8 standard development board containing 8MB flash; acts as local test master.
DL MCU	Door-Lock Microcontroller Unit; target slave controller executing lock, motor, and peripheral routines.
Camera MCU	Image validation module connected to the N8 via the SPI GPIO matrix; handles physical door status check.
MRDY / SRDY	Master Ready / Slave Ready; active-low physical lines gating UART serial transactions to prevent bus clashes.
DPID	Data Point Identifier; numeric command IDs used inside framing packets to select execution operations.
NVS / flash	Non-Volatile Storage key-value partition inside ESP32 SPI flash; retains Wi-Fi configs and firmware indices.
SoftAP	Software-enabled Access Point; local diagnostic wireless hotspot hosted by N8 for first-boot provisioning.
MQTT	Message Queuing Telemetry Transport; light publish/subscribe protocol routed over port 8883 (TLS).
QoS 1	Quality of Service level 1; guarantees at-least-once message delivery with acknowledgment tracking.

CHAPTER 2: DISTRIBUTED SYSTEM ARCHITECTURE & TIERS

2.1 Multi-Tiered System Architecture

The SIMLOCK test platform operates under a distributed four-tier architecture designed to isolate local safety-critical testing loops from network latency, while ensuring central fleet-wide consolidation of performance logs. Each N8 board operates as an autonomous testing agent: it compiles, translates, and drives test commands locally, then backhauls heartbeats, alerts, and results through the central MQTT broker to the persistent database server.

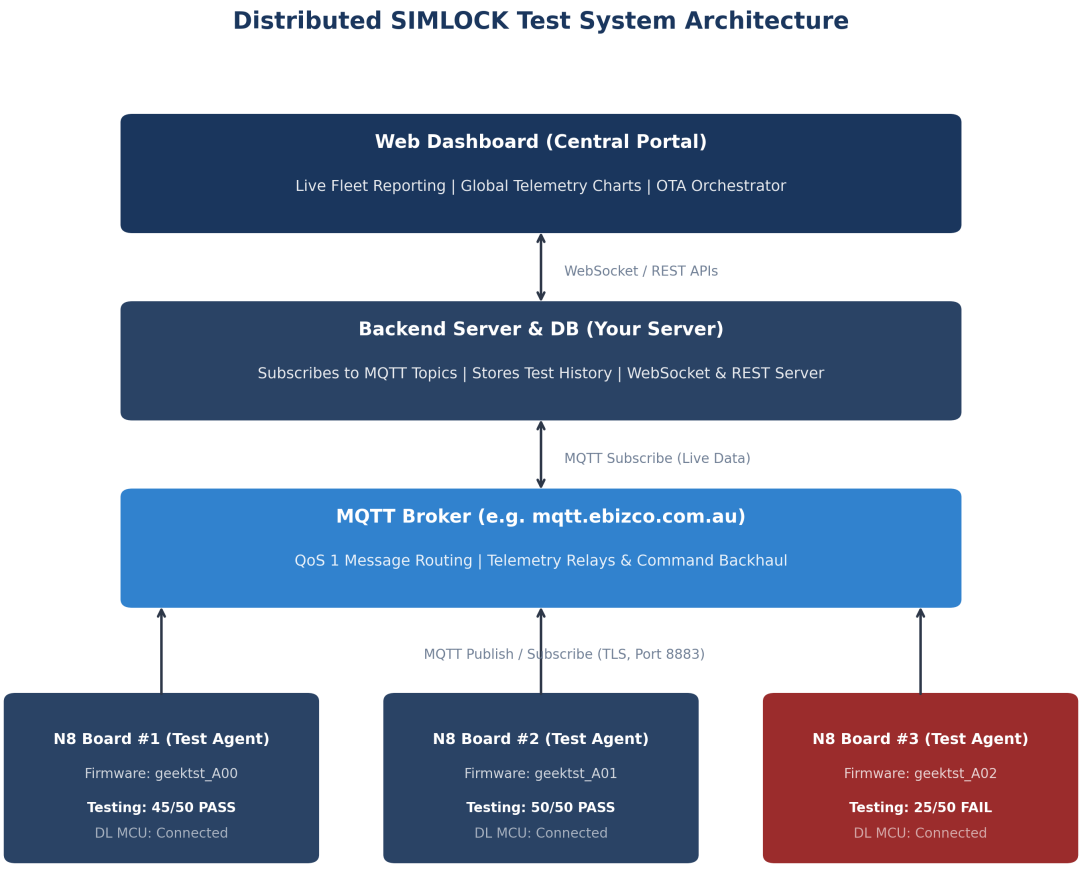


Figure 1: Distributed multi-tiered SIMLOCK testing network and broker relay.

This distributed layout establishes two completely independent interfaces on each local board: First, local on-bench control is achieved via REST API and WebSockets served natively by the N8's HTTP server over Wi-Fi. Second, multi-board logging is achieved by the N8 publishing JSON status packets through the designated broker to the central SQL database. This keeps diagnostic bench activities separate from high-level production tracking.

CHAPTER 3: N8 HARDWARE SPECIFICATIONS & PINOUTS

3.1 Peripheral Breakouts & Migration Rationale

The Waveshare ESP32-C6 dev board replaces the legacy Waveshare GEEK module. While the GEEK stick's five pins (GPIO 3, 4, 5, 6, 9) were sufficient to drive the basic DL MCU connection, they left no free pins to attach secondary camera verification systems. The N8 board breaks out the full silicon pin set, allowing the DL MCU interface and the new Camera MCU interface to operate concurrently without electrical overlaps.

N8 ↔ DL MCU 5-Wire Interconnect Pinout (Table 5)

Signal Name	N8 GPIO	DL MCU Connector Pin	Electrical Interface Direction
TXD	GPIO5	RX (MCU UART)	N8 Output ■■■ DL MCU Input
RXD	GPIO6	TX (MCU UART)	DL MCU Output ■■■ N8 Input
SRDY	GPIO3	SRDY (Active-Low)	DL MCU Output ■■■ N8 Input
MRDY	GPIO4	MRDY (Active-Low)	N8 Output ■■■ DL MCU Input
GND	GND	GND (Common)	Shared Reference Ground Line

N8 ↔ Camera MCU SPI Interconnect Pinout (Table 6)

Signal Name	N8 GPIO	Camera Pin	Electrical Interface Direction
SCK	GPIO19	SCK (Serial Clock)	N8 Output ■■■ Camera Module Input
MOSI	GPIO18	MOSI (Data Out)	N8 Output ■■■ Camera Module Input
MISO	GPIO20	MISO (Data In)	Camera Module Output ■■■ N8 Input
CS	GPIO21	CS (Chip Select)	N8 Output ■■■ Camera Module Input
3.3V	3.3V	VCC (V_System Source)	N8 Power Output ■■■ Camera Module Power
GND	GND	GND (Common)	Shared Reference Ground Line

3.2 Physical Silicon Constraints & Safety Cautions

CRITICAL ELECTRICAL WARNINGS — STRAPPING PINS AND JTAG CONFLICTS:

1. **Strapping Pins Complete Set:** Per Espressif's ESP32-C6 datasheet, five pins are sampled during power-on/reset: **GPIO4, 5, 8, 9, 15**. Since Table 5 assigns GPIO4 and GPIO5 to MRDY and TXD, the DL MCU must hold these lines in high-impedance state during N8 startup. On the N8 dev board, GPIO8 is hardwired to the RGB status LED and GPIO9 is hardwired to the BOOT button, leaving only GPIO15 as a free strapping pin.
2. **JTAG Signal Overlaps (GPIO4-7):** These pins comprise the hardware JTAG debugging interface (MTMS, MTDI, MTCK, MTDO). Three critical lock signals sit on this group (GPIO4/MRDY, GPIO5/TXD, GPIO6/RXD). While JTAG is disabled by default, developers must never enable on-chip JTAG debugging while the DL MCU is physically connected, as line contention will corrupt serial transactions.
3. **Camera SPI Matrix Routing Rationale:** FSPI IO_MUX pins (GPIO4, 5, 6) conflict with Table 5. Therefore, SPI signals must be routed to GPIO18-21 using the ESP32-C6 GPIO Matrix. This choice prevents pin overlaps but imposes a lower maximum SCLK frequency, which must be verified if frame rates exceed standard thresholds.
4. **Host Port Isolation:** GPIO12/13 (USB-C link) and GPIO16/17 (UART0 console logs) are kept clear of Tables 5/6.

N8 Dev Board physical & pinout interconnect map

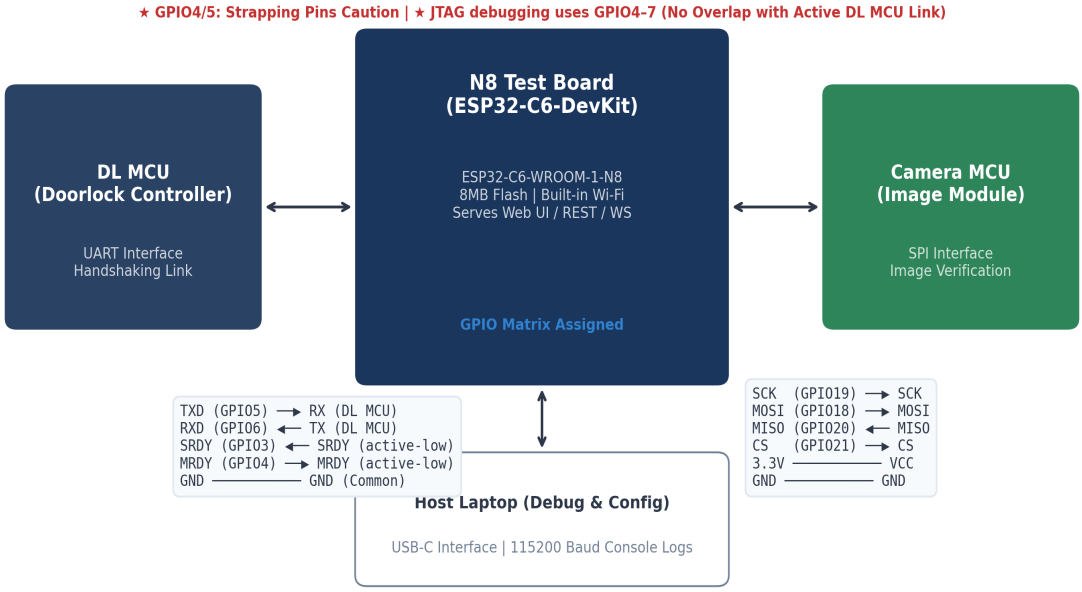


Figure 2: Hardware interconnect pinout mappings (DL MCU vs Camera MCU).

CHAPTER 4: COMMUNICATION PROTOCOL & COMMANDS

4.1 Active-Low Hardware Handshake Logic (MRDY/SRDY)

UART serial commands are gated by an active-low physical handshake between the N8 (Master) and the DL MCU (Slave). This prevents transmission buffer overruns and synchronization errors. The logic flow proceeds as follows: First, the N8 asserts MRDY low to request the bus. Second, the DL MCU acknowledges by asserting SRDY low. Third, the N8 transmits the command bytes over TXD. Fourth, the DL MCU replies over RXD. Fifth, both lines are released high (idle state).

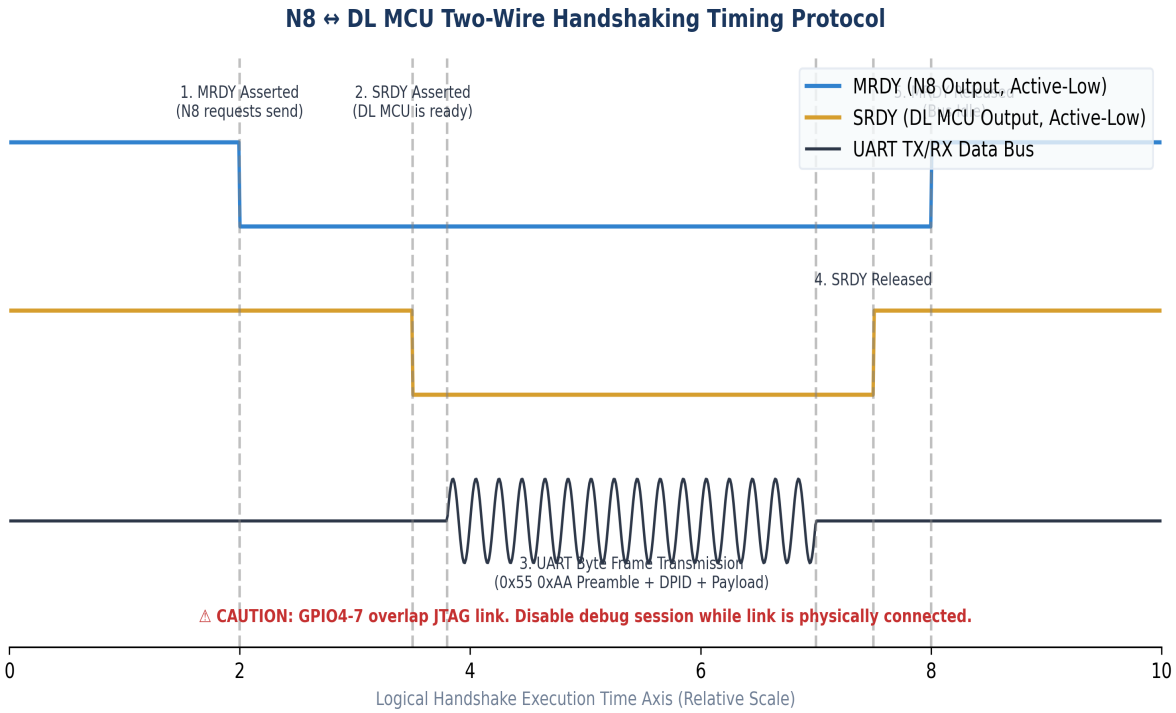


Figure 3: Two-wire active-low MRDY/SRDY serial handshaking timing protocol.

4.2 Command Frame Format & DPIDs

Each serial frame begins with the static two-byte preamble 0x55 0xAA, followed by an 8-bit Data Point Identifier (DPID), a length byte, and a variable-length payload. While v2.1 had unconfirmed commands, the v2.2 specification confirms core credential operations. Other system settings (LEDs, buzzers, and parameters) are routed using dedicated DPIDs as shown below.

DPID Value	Target Microcontroller Operation	Test Mapping & Scope
21	Add Credential (PIN / RFID / Fingerprint)	Confirmed in Tests 5, 6, 7, 13, 18. Supports durations.
22	Delete Credential by Hardware ID	Confirmed in Tests 8, 14, 19.
23	Buzzer Control (Beeps, Alarm)	Confirmed in Tests 22, 23, 24.
24	Backlight Controller (Flash, ON/OFF)	Confirmed in Tests 29, 30, 31.
25	Read System Status (RSSI, Battery)	Confirmed in Tests 34, 35, 36.
26	Auto-Lock Configuration	Confirmed in Tests 37, 38.

CHAPTER 5: DEVICE PROVISIONING & NETWORK INITIALIZATION

5.1 First-Boot Wireless Setup Sequence

When the N8 controller is powered on with no Wi-Fi credentials stored in flash NVS, it automatically spins up its internal access point (SoftAP) under the SSID 'ESP32-Setup-XXXX'. Users connect without a passcode to load the local gateway portal at http://192.168.4.1/. This portal captures local Wi-Fi settings and saves them to the NVS flash partition.

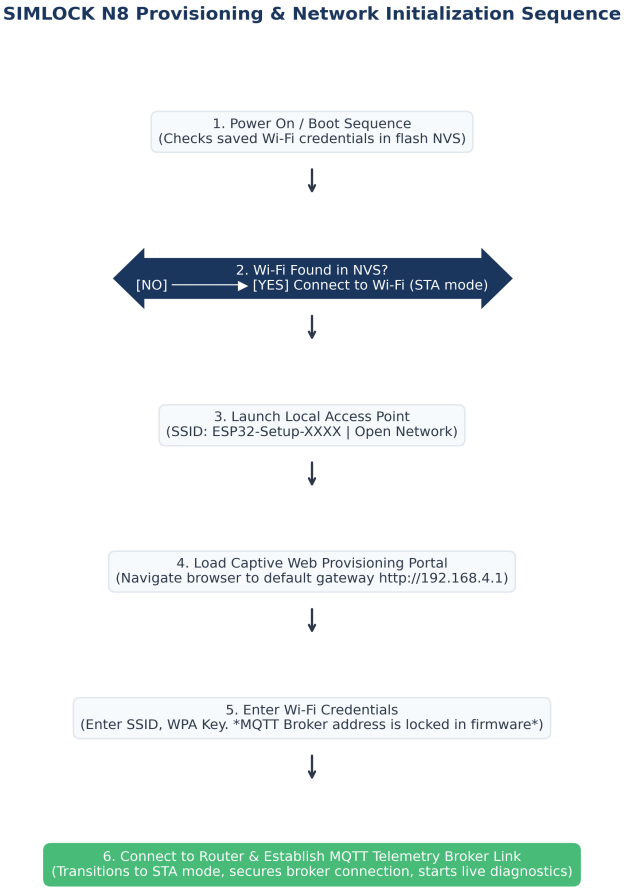


Figure 4: N8 provisioning flow: SoftAP + captive portal, followed by MQTT broker connection.

5.2 Modified Captive Portal UI (Auto-MQTT Routing)

UI/UX DESIGN UPDATE — REMOVAL OF MQTT CONFIGURATION OVERRIDES:

In this revised portal design (Figure 5), the MQTT broker parameter input has been completely removed from the captive portal configuration form. This ensures strict fleet alignment, preventing field operators from accidentally misconfiguring telemetry routes. The N8 test board automatically publishes and subscribes using pre-allocated broker hosts, ports, and QoS targets hardcoded securely inside the compiled firmware.



Figure 5: Captive provisioning web portal interface layout (No MQTT parameters exposed).

CHAPTER 6: LOCAL WEB PORTAL — CONFIGURATION SUITE EDITOR

6.1 Active Configuration Portal Page Layout

UI/UX LAYOUT UPDATE — COMPACT TEST EDITOR & SCROLLING SLIDING WINDOW:
After Wi-Fi setup, operators access the local SIMLOCK portal at http:///. To optimize workspace efficiency, the local Web UI (Figure 6) has been updated: First, the active test configuration title and the Test Event ID input field are merged into a single compact line. Second, buttons like 'Save CONFIG', 'import config', and 'Export JSON' are removed, since all test suites are ingested remotely. Third, the test list is rendered inside a scrollable window displaying at least 10 test configurations simultaneously, preserving vertical screen space for active debug monitors.

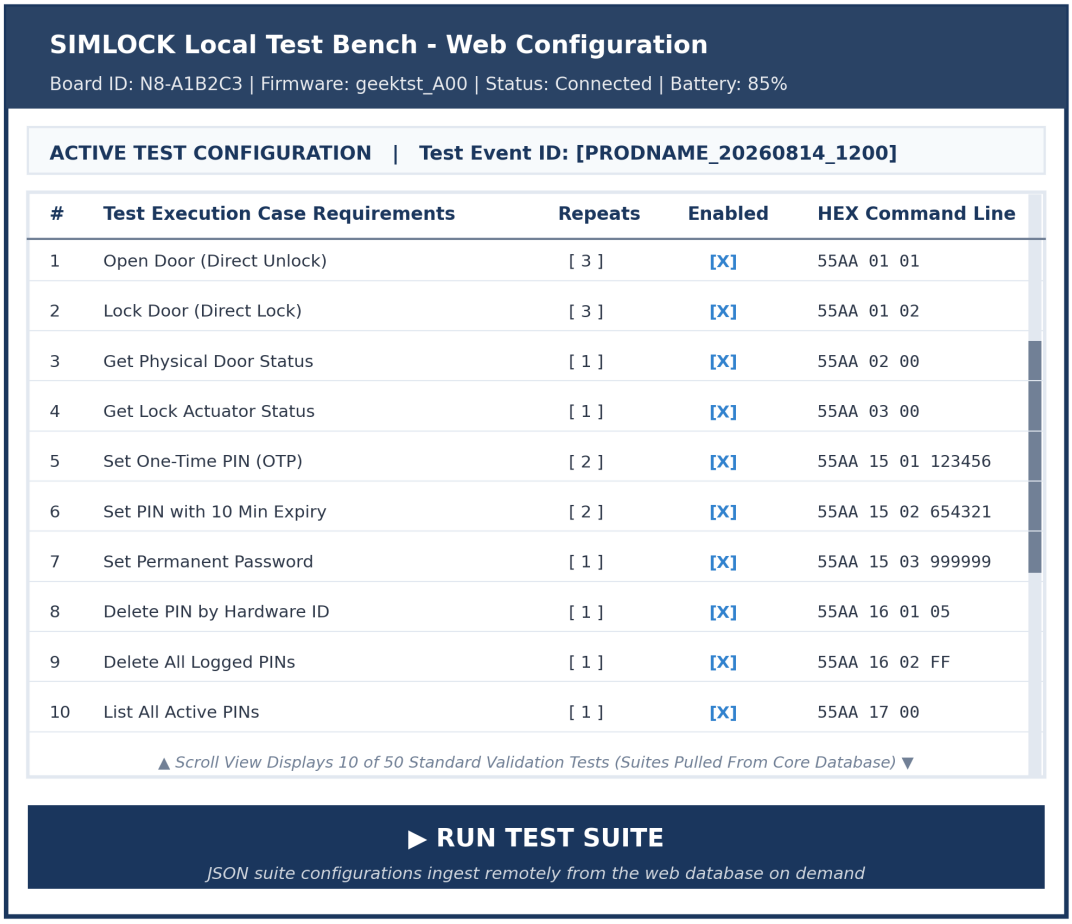


Figure 6: Local SIMLOCK browser-based web configuration portal (Scrolling list and merged header).

CHAPTER 7: REAL-TIME LIVE EXECUTION MONITOR

7.1 In-Flight Execution UI & Row Highlighting

UI/UX LAYOUT UPDATE — IN-FLIGHT CURRENT CASE HIGHLIGHT & EXPORT REMOVAL:
When an automated test run is triggered, the browser loads the Live Test Execution Monitor (Figure 7). To provide immediate visual context to bench technicians, the active row currently executing is highlighted using a light blue background fill (#EBF8FF) and bold border styling. This immediately draws the eye to the active test case. Additionally, the 'Export Execution Log' button has been removed from the local client interface. Because each test result is backhauled to the central server database in real time, local buffer exports are unnecessary, preventing disk fragmentation on the N8's internal SPIFFS partition.

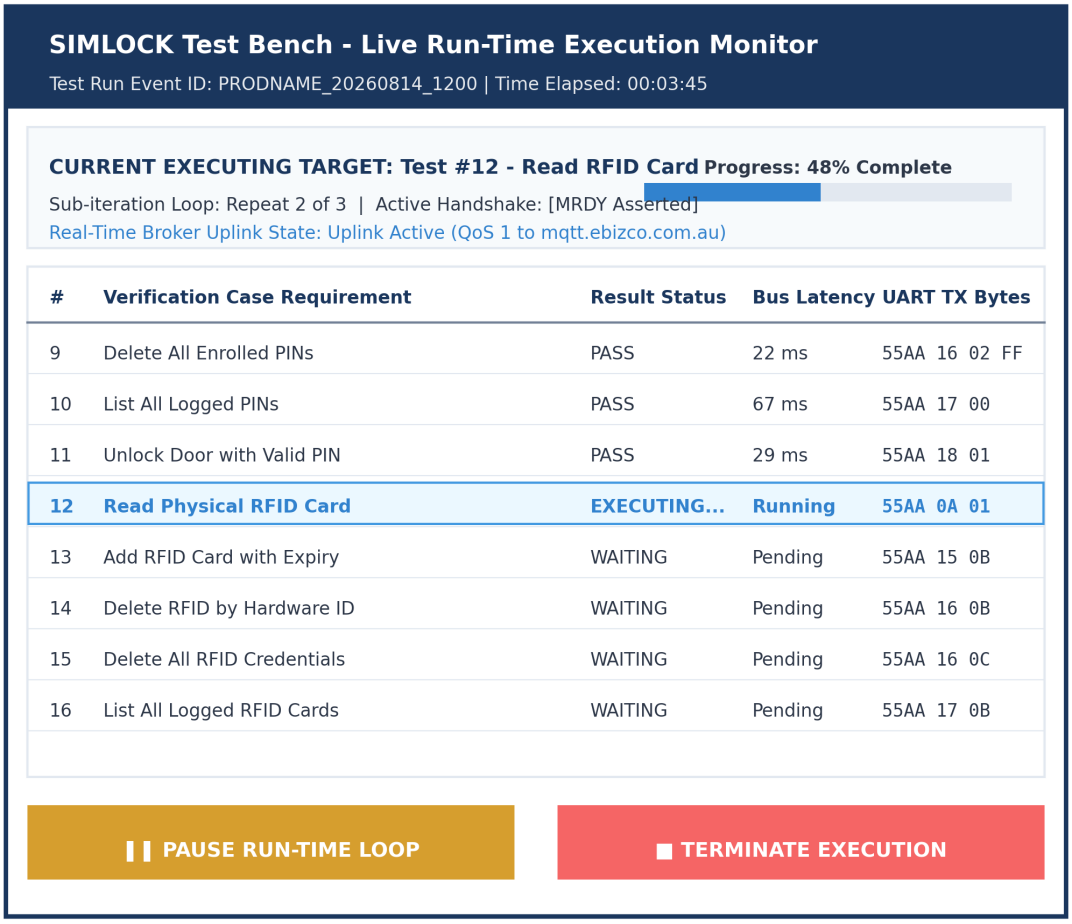


Figure 7: Local SIMLOCK live-execution run-time page (Highlighted row 12 under active test).

CHAPTER 8: CENTRALIZED CLOUD DASHBOARD & REMOTE VIEWING

8.1 Cloud Remote Viewing Page & Database Logging Portal

NEW SYSTEM CAPABILITY — BROADCAST FLEET DB AND REMOTE WORKSTATIONS:
Because test results backhaul automatically via MQTT, technicians monitor active test stands remotely using the central database monitoring page (Figure 8). The Centralized Cloud Dashboard subscribes to telemetry topics from all deployed N8 test controllers and parses them into a live-updating webpage logging database. This remote portal consolidates: First, a Fleet Operational Summary panel displaying pass rates, log counts, and connection health. Second, an Active Testing Agent Grid displaying online firmware revisions and loop progress. Third, a scrollable Database Telemetry Stream log displaying raw UART status bytes and physical latch response latency from remote workstations.

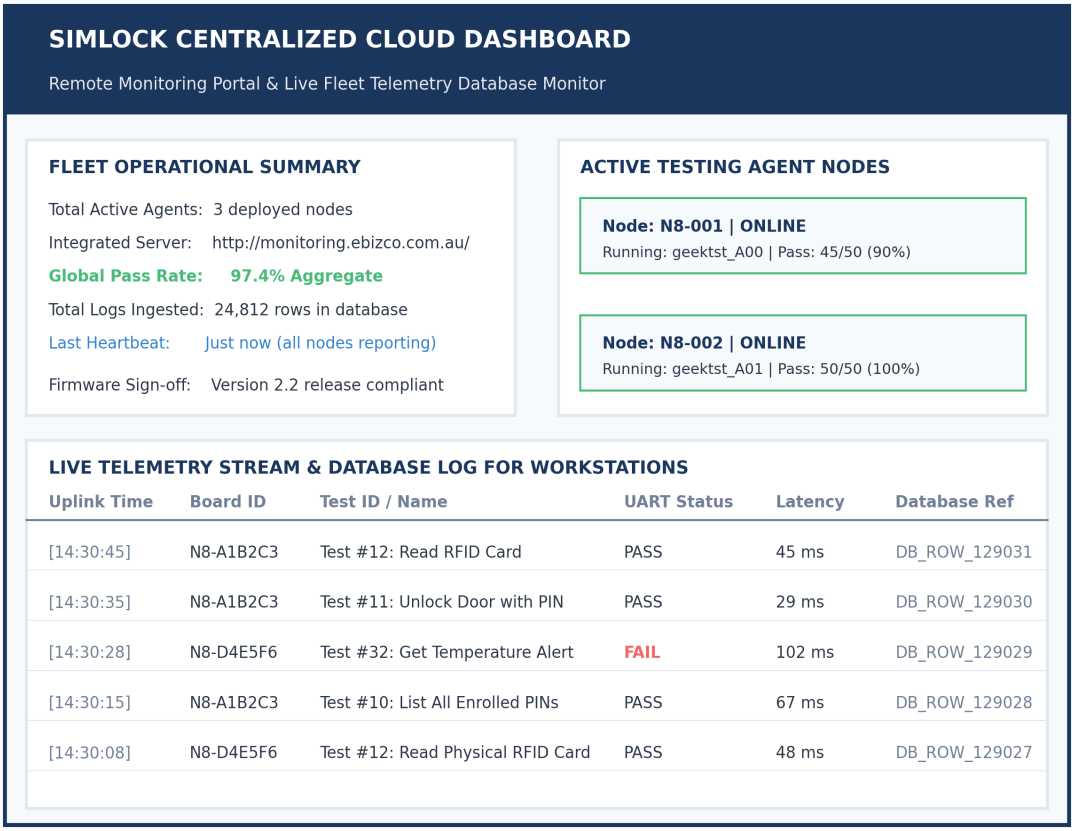
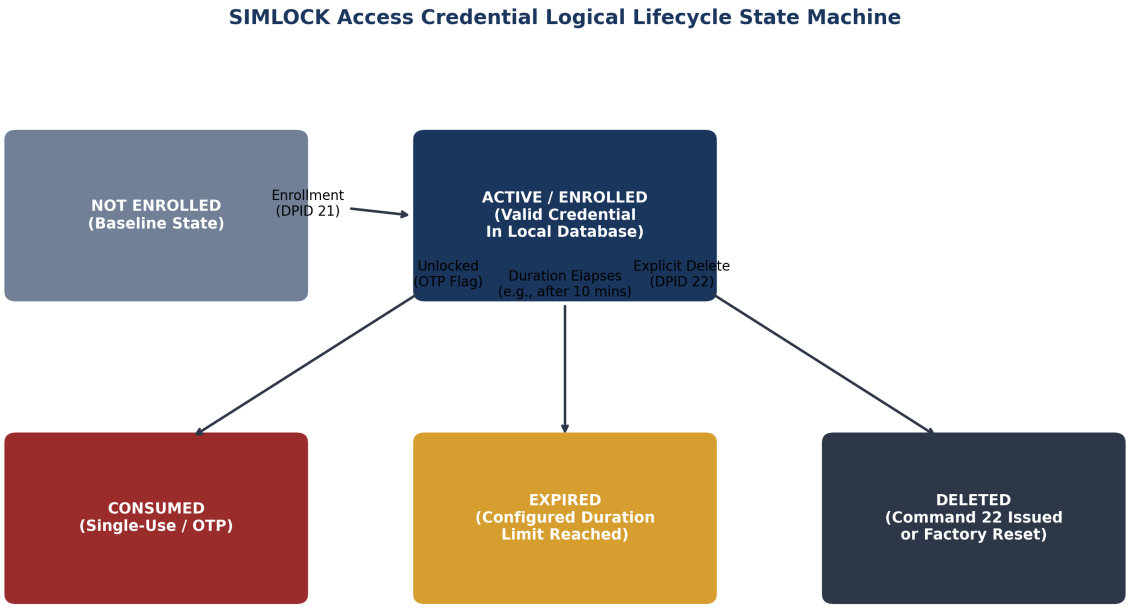


Figure 8: SIMLOCK Cloud Fleet Remote Viewing Page (Central monitoring dashboard and DB log stream).

CHAPTER 9: CREDENTIAL LOGICAL LIFECYCLE MODEL

9.1 State Transitions of Enrolled Access Credentials

Access credentials (PIN, RFID, and fingerprint templates) share a common lifecycle state machine. A credential is added via DPID 21, transitioning from 'Not Enrolled' to the 'Active' state. Depending on its configuration, it follows one of three terminal paths: One-time credentials (such as OTP PINs) become 'Consumed' upon first unlock. Duration-limited credentials automatically transition to 'Expired' when their configured lifespan elapses. Any credential transitions to 'Deleted' upon an explicit delete command (DPID 22) or physical boot reset.



Note: Consumed, Expired, and Deleted are Terminal States. Physical door locks will reject credentials.

Figure 9: Credential lifecycle shared by PIN, RFID, and fingerprint credentials.

CHAPTER 10: MQTT FLEET REPORTING TOPICS & DATA MAPPINGS

10.1 Telemetry Broadcast Topic Mapping Structures

N8 controllers establish persistent TCP links to the preconfigured MQTT broker on port 8883 (TLS). Upon connection, the controller publishes heartbeat, status, test-result, and system-level trace logs. Each topic uses the deterministic physical MAC address client-id format as shown in Table 11.1.

Broker Topic Structure	Format	Publish Frequency	Functional Description & Payload Scope
test/boards/{id}/status	JSON	Every 10 seconds	Heartbeat packet: Wi-Fi RSSI, local IP, uptime, and battery level.
test/boards/{id}/test/start	JSON	On execution launch	Metadata header payload: unique Test Event ID and case total counts.
test/boards/{id}/test/result	JSON	On test completion	Individual result tracking: PASS/FAIL status, response latencies, and HEX traces.
test/boards/{id}/test/complete	JSON	On suite completion	Aggregate summary packet: totals passed, failed, skipped, and durations.
test/boards/{id}/log	JSON	As system events occur	Low-level system traces and diagnostic serial messages.
test/boards/{id}/alert	JSON	As alerts are triggered	Security alert packets: failed finger-sweeps, stuck bolts, high temperatures.
test/boards/{id}/ota/status	JSON	During active updates	OTA update status indicators: current, target versions, and percentage indices.

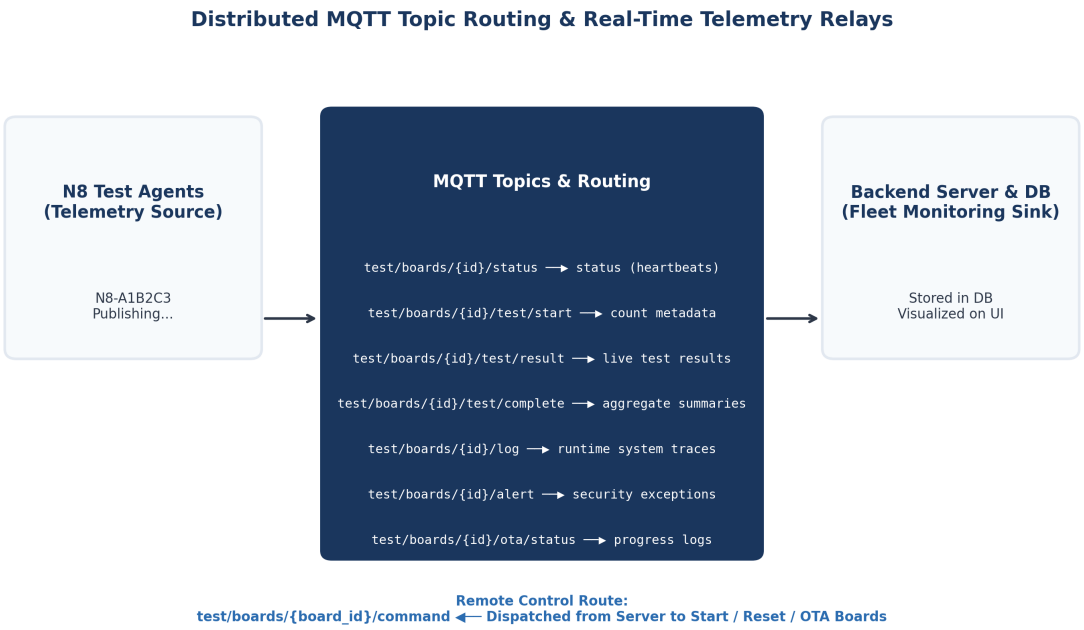


Figure 10: MQTT telemetry publish and subscribe command routing structures.

CHAPTER 11: PORTABLE TEST SUITE DEFINITION JSON SCHEMA

11.1 Dynamic Test Definition Schema Configuration

Test suites are compiled and defined in portable, easily-edited JSON schemas. These JSON definitions are prepared remotely and pushed over MQTT to override current N8 on-board structures. This removes the need for local Web UI file uploads and manual configuration saves. A baseline JSON schema mapping direct control and credential parameters is defined below.

```
{
  "version": "1.0",
  "product_name": "OZKEY",
  "test_definitions": [
    {
      "test_id": 1,
      "test_name": "Open Door (Direct Unlock)",
      "section": "Direct Control",
      "enabled": true,
      "repeats": 3,
      "command": {
        "dpid": 21,
        "payload_hex": "01",
        "description": "Unlock command"
      },
      "expected_response": {
        "dpid": 21,
        "status": "00",
        "payload_hex": "00"
      },
      "timeout_ms": 10000,
      "delay_between_tests_ms": 10000
    }
  ]
}
```

11.2 Run-Time Fields Modifiability Matrix

JSON Schema Target Field Path	Modifiability State	Functional Modification Purpose
command.dpid	Editable	Alters active DPID command selectors if hardware protocols update.
command.payload_hex	Editable	Modifies raw hex parameters sent across UART TX wires.
expected_response.status	Editable	Alters return success checks matching validation baselines.
timeout_ms / delay_between_tests_ms	Editable	Increases wait limits and inter-case cool-down spacings.
repeats / enabled	Editable	Toggles cases in and out of active run arrays; sets stress repeat cycles.
test_id	Read-Only	Immutable static database index (1 to 50) of verification requirements.
test_name / section	Read-Only	Immutable requirement titles and static functional matrix categories.

CHAPTER 12: COMPREHENSIVE LOCAL REST & WEBSOCKET ENDPOINTS

12.1 Local REST Web Service Catalog Reference (All 40 Endpoints)

Each N8 controller hosts a local web service on port 80. While the MQTT broker is used for fleet Backhauling, local REST and WebSocket interfaces are used for single-stand direct bench debugging. These endpoints cover all 11 functional modules of the system specifications, as fully cataloged below.

Method	REST API Endpoint Path	Functional Diagnostic Purpose / Description
GET	/api/version	Get current firmware version string (e.g. geektst_A00)
GET	/api/status/battery	Retrieve local battery status and capacity percent
GET	/api/status/rssi	Get connected Wi-Fi Received Signal Strength Indicator (RSSI) in dBm
POST	/api/lock/unlock	Send manual unlock instruction frame to DL MCU over UART
POST	/api/lock/lock	Send manual lock instruction frame to DL MCU over UART
GET	/api/lock/status	Query current physical latch and deadbolt locking state
POST	/api/credential/pin/otp	Enrolls single-use PIN (OTP) using command DPID 21
POST	/api/credential/pin/duration	Adds temporary passcode with auto-expiry lifetime parameters
POST	/api/credential/pin/permanent	Registers indefinite, non-expiring master passcodes in database
DELETE	/api/credential/pin/{id}	Deletes specific passcode identified by Hardware ID from DL MCU database
DELETE	/api/credential/pin/all	Triggers bulk-delete instruction clearing passcodes from lock
GET	/api/credential/pin/list	Returns complete manifest of registered passcodes, exps, and IDs

Method	REST API Endpoint Path	Functional Diagnostic Purpose / Description
POST	/api/credential/rfid/read	Enters card-reading listen loop; polls DL MCU reader
POST	/api/credential/rfid/duration	Registers temporary physical RFID card credential with timeout bounds
DELETE	/api/credential/rfid/{id}	Removes card record mapping to specified target Hardware ID
DELETE	/api/credential/rfid/all	Clears all physical card records from lock credential DB
GET	/api/credential/rfid/list	Returns complete catalog of all enrolled active RFID cards
POST	/api/credential/finger/read	Enters finger scanner loop; polls physical sensor sweeps
POST	/api/credential/finger/duration	Enrolls fingerprint templates restricted by temporary lease spans
DELETE	/api/credential/finger/{id}	Removes fingerprint template mapped to specified target Hardware ID
DELETE	/api/credential/finger/all	Triggers bulk wipe of all fingerprint templates from storage
GET	/api/credential/finger/list	Returns index listing for all active enrolled fingerprint IDs
POST	/api/buzzer/beep	Causes DL MCU on-board buzzer to sound a single brief beep
POST	/api/buzzer/double	Dispatches sequence causing buzzer to issue double beeps
POST	/api/buzzer/alarm	Triggers acoustic emergency alarm pattern on DL MCU
POST	/api/alert/failed-attempt	Fires test logging trigger simulating unauthorized finger sweeps
POST	/api/alert/stuck-bolt	Simulates mechanical lock actuator stuck bolt warning events
POST	/api/alert/high-temp	Simulates high temperature sensor warnings (>70C boundary condition)
POST	/api/alert/duress	Triggers quiet duress silent telemetry alert across MQTT broker
POST	/api/backlight/flash	Causes keypad backlight LEDs to flash exactly twice on lock

Method	REST API Endpoint Path	Functional Diagnostic Purpose / Description
POST	/api/backlight/on	Sets keypad backlight LED group to permanent illuminated state
POST	/api/backlight/off	Powers down keypad backlight LED groups entirely
GET	/api/log/transaction	Retrieves diagnostic unlock logs buffer stored inside RAM cache
GET	/api/log/alert	Reads chronological alert transaction history from flash database
POST	/api/system/reboot	Triggers hard reboot of local ESP32-C6 controller via software resets
POST	/api/system/factory-reset	Wipes flash NVS, clears credentials, and activates SoftAP AP
GET	/api/system/info	Returns manufacturer, hardware serial numbers, and platform revisions
GET	/api/credential/list/all	Queries combined manifest of active hardware IDs for debugging
GET	/api/log/export	Provides raw transactional logs export format (CSV or JSON)
POST	/api/ota/check	Checks remote update-server manifest for latest binaries available
GET	/api/ota/status	Queries current downloading or flashing progress percentage of updates
POST	/api/ota/update	Triggers immediate firmware download and flashing of remote binary
POST	/api/ota/rollback	Restores previous executable firmware image backed up in flash partition
POST	/api/ota/trigger	Simulates physical boot-button triple click to force OTA run
GET	/ws	WebSocket connection path routing real-time on-bench status updates

CHAPTER 13: REQUIREMENTS TRACEABILITY MATRIX & APPROVALS

13.1 Requirements Test-to-Section Traceability Matrix (Tests 1 to 50)

To support rigorous validation planning, Table 13.1 maps every validation case from the original 50-case SIMLOCK GEEK test procedure directly to its logical implementation section in this unified manual.

Test Case ID	Test Case Description Summary	Unified Specification Section Mapping
Test 1	Open Door / Unlock Latch	Section 6.1 (Table 12) & REST Endpoints (Chapter 12)
Test 2	Lock Door / Lock Latch	Section 6.1 (Table 12) & REST Endpoints (Chapter 12)
Test 3	Get Latch Physical Status	Section 6.1 (Table 12) & REST Endpoints (Chapter 12)
Test 4	Get Lock Actuator Status	Section 6.1 (Table 12) & REST Endpoints (Chapter 12)
Test 5	Set One-Time PIN (OTP)	Section 6.2 (Table 13) & Credential Lifecycle (Chapter 9)
Test 6	Set PIN with Expiry Duration	Section 6.2 (Table 13) & Credential Lifecycle (Chapter 9)
Test 7	Set Permanent Password	Section 6.2 (Table 13) & Credential Lifecycle (Chapter 9)
Test 8	Delete PIN by Hardware ID	Section 6.2 (Table 13) & REST Endpoints (Chapter 12)
Test 9	Delete All Enrolled PINs	Section 6.2 (Table 13) & REST Endpoints (Chapter 12)
Test 10	List All Enrolled PINs	Section 6.2 (Table 13) & REST Endpoints (Chapter 12)
Test 11	Unlock with PIN (Web UI)	Section 6.2 (Table 13) & REST Endpoints (Chapter 12)
Test 12	Read RFID Card Identifier	Section 6.3 (Table 14) & REST Endpoints (Chapter 12)
Test 13	Add RFID Card with Duration	Section 6.3 (Table 14) & Credential Lifecycle (Chapter 9)
Test 14	Delete RFID by Hardware ID	Section 6.3 (Table 14) & REST Endpoints (Chapter 12)
Test 15	Delete All Enrolled RFID Cards	Section 6.3 (Table 14) & REST Endpoints (Chapter 12)
Test 16	List All Registered Cards	Section 6.3 (Table 14) & REST Endpoints (Chapter 12)

Test Case ID	Test Case Description Summary	Unified Specification Section Mapping
Test 17	Read Fingerprint Sweeps	Section 6.4 (Table 15) & REST Endpoints (Chapter 12)
Test 18	Add Fingerprint with Duration	Section 6.4 (Table 15) & Credential Lifecycle (Chapter 9)
Test 19	Delete Finger by Hardware ID	Section 6.4 (Table 15) & REST Endpoints (Chapter 12)
Test 20	Delete All Fingerprints	Section 6.4 (Table 15) & REST Endpoints (Chapter 12)
Test 21	List All Registered Fingers	Section 6.4 (Table 15) & REST Endpoints (Chapter 12)
Test 22	Buzzer: Single Beep Command	Section 6.6 (Table 16) & REST Endpoints (Chapter 12)
Test 23	Buzzer: Double Beep Command	Section 6.6 (Table 16) & REST Endpoints (Chapter 12)
Test 24	Buzzer: Alarm Pattern Command	Section 6.6 (Table 16) & REST Endpoints (Chapter 12)
Test 25	Failed-Attempt Alert Log	Section 6.6 (Table 16) & REST Endpoints (Chapter 12)
Test 26	Stuck Bolt Mechanical Alert Log	Section 6.6 (Table 16) & REST Endpoints (Chapter 12)
Test 27	High Temperature Alert Log	Section 6.6 (Table 16) & REST Endpoints (Chapter 12)
Test 28	Silent Duress Alarm Log	Section 6.6 (Table 16) & REST Endpoints (Chapter 12)
Test 29	Backlight: Flash 2 Times	Section 6.7 (Table 17) & REST Endpoints (Chapter 12)
Test 30	Backlight: Permanent ON State	Section 6.7 (Table 17) & REST Endpoints (Chapter 12)
Test 31	Backlight: OFF State	Section 6.7 (Table 17) & REST Endpoints (Chapter 12)
Test 32	Read Transaction Logs Cache	Section 6.8 (Table 18) & REST Endpoints (Chapter 12)
Test 33	Read Alert Event Logs Cache	Section 6.8 (Table 18) & REST Endpoints (Chapter 12)

Test Case ID	Test Case Description Summary	Unified Specification Section Mapping
Test 34	Get Battery Level Percent	Section 6.8 (Table 18) & REST Endpoints (Chapter 12)
Test 35	Get Wi-Fi RSSI Signal dBm	Section 6.8 (Table 18) & REST Endpoints (Chapter 12)
Test 36	Get Current Firmware Version	Section 6.8 (Table 18) & REST Endpoints (Chapter 12)
Test 37	Configure Auto-Lock Delay Timer	Section 6.9 (Table 19) & REST Endpoints (Chapter 12)
Test 38	Enable / Disable Auto-Lock	Section 6.9 (Table 19) & REST Endpoints (Chapter 12)
Test 39	Configure Wrong PIN Lockouts	Section 6.9 (Table 19) & REST Endpoints (Chapter 12)
Test 40	Configure Scramble Passcodes	Section 6.9 (Table 19) & REST Endpoints (Chapter 12)
Test 41	Configure Buzzer Volume Decibels	Section 6.9 (Table 19) & REST Endpoints (Chapter 12)
Test 42	Configure Keypad Language Set	Section 6.9 (Table 19) & REST Endpoints (Chapter 12)
Test 43	Consolidated Hardware ID List	Section 6.9 (Table 19) & REST Endpoints (Chapter 12)
Test 44	Factory Reset Trigger (REST)	Section 6.10 (Table 20) & REST Endpoints (Chapter 12)
Test 45	Reboot Device Controller (REST)	Section 6.10 (Table 20) & REST Endpoints (Chapter 12)
Test 46	Query Manufacturer Device Info	Section 6.10 (Table 20) & REST Endpoints (Chapter 12)
Test 47	Logs Buffer Diagnostics Export	Section 6.10 (Table 20) & REST Endpoints (Chapter 12)
Test 48	Version Cycling (Triple-Click)	Section 6.11 (Table 21) & OTA Section (Chapter 11)
Test 49	OTA Update Check Trigger	Section 6.11 (Table 21) & REST Endpoints (Chapter 12)
Test 50	Boot Button Triple-Click NVS	Section 6.11 (Table 21) & REST Endpoints (Chapter 12)

13.2 Appendices & Approvals

Appendix A: Reference Test Environment Record

Capture physical test-bench parameters below to document compliance audits:

Audit Tracking Parameter	Recorded Bench Value / Sign-off
N8 Dev Board Hardware ID Code	_____
N8 Controller Boot Firmware Version	_____
Local Wi-Fi Router SSID Reference	_____
N8 Station Network IP Address	_____
Central Telemetry MQTT Broker Link Status	_____
Date and Time of Field Audit	_____
Assigned Validation Technician Name	_____

Appendix B: Formal Engineering Release Sign-offs

This technical manual and operational guide is approved for platform validation releases:

Assigned Release Role	Lead Authority Signature	Date Signed Compliant
Lead Test Engineer	_____	____ / ____ / 2026
Hardware Engineering Lead	_____	____ / ____ / 2026
Firmware Development Lead	_____	____ / ____ / 2026
Backend & Cloud Operations Principal	_____	____ / ____ / 2026